

Milestone 1 Report

DL & GenAI Project – Data Preprocessing and Exploratory Analysis

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Introduction

This milestone focuses on preprocessing and exploratory analysis of the Messy Mashup – Genres Stems Dataset.

The dataset consists of multiple genres, each containing individual song folders with four stems:

- *drums.wav*
- *vocals.wav*
- *bass.wav*
- *other.wav*

The objective was to:

- *Validate dataset integrity*
- *Identify corrupted or inconsistent files*
- *Perform silence analysis*
- *Create train-validation splits*
- *Perform basic audio mixing analysis*

Dataset Construction and Validation

A custom `build_dataset()` function was implemented to:

- *Validate presence of all 4 stems*
- *Check corrupted files (very small file sizes)*
- *Detect abnormal file size distribution*
- *Split dataset into training and validation sets (83% / 17%)*

Results

Q1 - Corrupted + Small Files

Total corrupted or unusually small files:

Ans: 1256

Q2 - File Size Imbalance

Absolute difference between large and small files:

Ans: 1072

Q3 - Train/Validation Split Difference

Absolute difference between:

- Reggae train drums
- Country validation vocals

Result:

Ans: 66

Silence Analysis

A function `find_long_silences()` was implemented using:

- `librosa.load`
- `librosa.effects.split`
- Silence threshold = 5 seconds
- `top_db = 20`

The analysis identified silence location:

- Start
- Middle

- *End*
- *Full silence*

Silence Analysis Results

Q4 - Total files with silence ≥ 5 sec

Ans: 680

Q5 - Vocals files with silence ≥ 5 sec

Ans: 304

Q6 - Average silence duration (vocals)

Ans: 12.59 seconds

Key point: This suggests vocals contain significant long silence segments.

Q7 - Jazz drums silence ≥ 5 sec

Ans: 24

Q8 - Jazz drums with only middle silence

Ans: 0

Key point: No purely middle-only silence detected in jazz drums.

Q9 - Jazz drums silence ≥ 10 sec

Ans: 7

Key point: This indicates some extreme silence cases in jazz drum stems.

Audio Mixing Analysis

To analyze mixing properties:

- *First rock song selected*
- *All 4 stems loaded*
- *Summed to create raw mix*

Mixing Results

Q10 - Mix Length

Ans: 110250 samples

Key point: (5 seconds at 22050 Hz)

Q11 - RMS Value

Ans: 0.20

Key point: Indicates moderate signal energy.

Q12 - Maximum Peak Before Normalization

Ans: 0.96006984

Key point: Since peak < 1.0, clipping does not occur before normalization.

Observations

- Large number of small/corrupted files detected (1256)
- Silence is common in vocals stems
- Jazz drums contain some long silence cases
- File size imbalance exists
- Raw mix does not clip but is close to peak threshold

Conclusion

This milestone successfully:

- Cleaned and structured dataset
- Identified corrupted and inconsistent files
- Performed silence pattern analysis
- Verified train-validation split balance
- Evaluated basic mixing characteristics

The dataset is now prepared for feature engineering and model training in the next milestone.

